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**NON-LINEAR PROGRAMMING FOR ASSEMBLY LINE BALANCING WITH
MULTI-MAN STATION AND PARALLEL TASK ASSIGNMENT AND A CASE OF
HOME APPLIANCES MANUFACTURING FACTORY**

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TEŞEKKÜRLER

Öncelikle bu zorlu pandemi sürecine rağmen tez süresince her sorunumuzda yanımızda olan, bizlere desteğini eksiltmeyen, tecrübesi ve donanımıyla hem akademik hem de insanlık olarak bize yol gösteren, tezin ilerlemesinde ve tamamlanmasında kritik bir rol oynamış değerli danışman hocamız Sn. Prof. Dr. Alev Taşkın GÜMÜŞ’e teşekkürlerimizi sunarız.

Bizleri bugünlere getiren hayattaki tüm sıkıntılarla ve zorluklarla mücadele etmeyi öğreten, hiçbir zaman desteklerini esirgemeyen, daima arkamızda olduklarının güveniyle hayatta ilerlememizi sağlayan, ailemize, her zaman bize inandıkları ve bizim istediğimiz şeyleri, bizim istediğimiz şekilde yapabileceğimiz bir ortam oluşturdukları için sonsuz teşekkür ederiz.

Büşra KARAALİ & İlknur SEZEN & Mustafa KURTOĞLU

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LIST OF SYMBOLS

<i>Symbol</i>	<i>Meaning</i>
M	Number of task
N	Number of station
O	Number of operator
t_i	Processing time of task i
P_i	Presedence of task i
L_k	Presedence of operator k
U	Upper-limit for parallel tasks
R	Upper-limit for multi-manned stations
X_{ij}	$\begin{cases} 1, \text{ If task } i \text{ is assigned to station } j \\ 0, \text{ otherwise} \end{cases}$
Y_i	$\begin{cases} 1, \text{ If task } i \text{ is assigned to multiple stations} \\ 0, \text{ otherwise} \end{cases}$
W_{kj}	$\begin{cases} 1, \text{ If operator } k \text{ is assigned to station } j \\ 0, \text{ otherwise} \end{cases}$
V_j	$\begin{cases} 1, \text{ If multiple operators are assigned to station } j \\ 0, \text{ otherwise} \end{cases}$
C	Cycle time

Z	Objective Function
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LIST OF ABBREVIATIONS

SALBP: Simple Assembly Line Balancing Problem

GALBP: Generalized Assembly Line Balancing Problem

ALBP: Assembly Line Balancing Problem

MALBP: Mixed-model Assembly Line Balancing Problems

ALWABP: Assembly Line Worker Assignment and Balancing Problem

TALBP: Two-Sided Assembly Line Balancing Problem

RMALB/S: Robotic Mixed-Model Assembly Lines

(MuRALB): Robotic Assembly Line with Multiple Models Problem

(MILP): Mixed Integer Linear Programming

RALB: Robotic Assembly Line balancing

SULBP: Simple U-Line Balancing Problem

UALBP: U -Line Balancing Problem

SMD: Single Model Deterministic

SMS: Single Model Stochastic

MMD: Multi/Mixed Model Deterministic

MMS: Multi/Mixed Model Stochastic

MSP: Mixed Model Sequencing Problem

VSM: Value-Stream Mapping

MRP: Material Requirements Planning

SD: Single Door

BM: Bottom Mount

PDM: Precedence Diagramming Method

GA: Genetic Algorithm

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ABSTRACT

Assembly lines are special production flow-line systems that are of great importance in the industrial production of high quantity products. And as a result of increase in demand brought by the increasing population, supported by developing technology, the value given to productivity and the correct usage of resources has increased in manufacturing sector. With the sector being more competitive than ever optimization of the factory, balancing of each line has become critical.

For decades, line balancing has attracted the attention of researchers trying to find solutions to its real world applications. While tremendous work has been done in this field, there is still a gap between research and real problems. After evaluating over 100 articles, this study provides a new non-linear mixed integer mathematical programming approach to solve the line balancing problem that has been provided by one of the leading companies in manufacturing sector particularly at home appliances. Applying model to given data, which is a SALBP-2 in this case, demonstrates the gains of the parallel line assignment and multi-manned workstation approach in line balancing studies. After that, difficulties with Mathematical Programming Based Approaches are discussed and a framework is proposed for future studies which contains suggestion of using Genetic Algorithms from Meta-Heuristics Approach.

Keywords: Value flow mapping, precedence diagram, SALBP-2, mixed integer programming, non-linear modeling, parallel task assignment

ÖZET

Montaj hatları, seri imalat ürünlerin endüstriyel üretiminde büyük önem taşıyan özel üretim akış hattı sistemleridir. Gelişen teknoloji ile desteklenen, artan nüfusun getirdiği talep artışı sonucunda, üretim sektöründe verimliliğe ve kaynakların doğru kullanımına verilen değer artmıştır. Sektördeki rekabetin zirveye çıkması ile fabrikanın iyi bir şekilde optimize edilmesi ve, her hattın dengelenmesi kritik hale gelmiştir.

On yıllardır hat dengeleme, gerçek dünya uygulamalarına çözüm bulmaya çalışan araştırmacıların dikkatini çekmiştir. Bu alanda oldukça fazla çalışma yapılsa da, yapılan araştırmalar ile gerçek sorunlar arasında hala bir boşluk var. 100'den fazla makaleyi değerlendirdikten sonra, bu çalışma özellikle ev aletlerinde imalat sektörünün önde gelen şirketlerinden biri tarafından sağlanan hat dengeleme problemini çözmek için lineer olmayan yeni bir matematiksel programlama yaklaşımı sunmaktadır. Bu durumda, basit hat dengeleme problemi olan eldeki verilere modelin uygulanması, hat dengeleme çalışmalarında, paralel görev ataması ve çok insanlı iş istasyonu yaklaşımının kazanımlarını göstermektedir. Daha sonrasında ise, Matematiksel Programlamaya Dayalı Yaklaşımlar ile ilgili zorluklar tartışılmakta ve Meta-Sezgisel Yaklaşım'dan Genetik Algoritmaların kullanılmasını öneren gelecekteki çalışmalar için bir çerçeve sunulmaktadır.

Anahtar Kelimeler: Değer akışı haritalama, öncelik diyagramı, MHDP, lineer olmayan modelleme, paralel görev atama yaklaşımı

1. INTRODUCTION

CHAPTER I

1.1 Literature Review

We researched studies on assembly line balancing between 2015-2020. These studies are summarized below by year. The classified table of the literature review can be seen in acknowledgement 1.

The researches conducted in 2015 are as follows:

Markus Pröpster, Lothar März, Gunther Reinhart and Carsten Intra introduced an approach which validates line balancing results by using the simulation tool [1]. To minimize the expected cycle time under the presence of uncertain workers by Marcus Ritt, Alysson M. Costa and Cristóbal Miralles, the assembly line is modeled as a two-stage mixed integer program to solve the problem of worker assignment and balancing, and local search heuristics is proposed to solve [2]. Lorenzo Tiacchi first addressed the buffer allocation problem and assembly line balancing problem (ALBP) at the same time in 2015 [3]. He offered an innovative approach, combining the latest developments of simulation techniques with a genetic algorithm. Hugo Zupan and Niko Herakovic presented a case study using balancing and discrete event simulation approach for optimization of the production line [4]. Reza Ramezani and Abdullah Ezzatpanah solved mixed-model assembly line balancing and worker assignment problem with an imperialist competitive algorithm [5]. Manuel Chica, Óscar Cerdón, Sergio Damas and Joaquín Bautista proposed an interactive multicriteria optimization framework for the time and space assembly line balancing problem [6]. It was presented a line balancing and scheduling methodology for the industrialization stage in Airbus by T.Borreguero, F.Mas, J.L.Menéndez and M.A.Barreda in [7]. An innovative approach to deal with the Mixed Model Assembly Line Balancing Problem (MALBP) with stochastic task times and

parallel workstations is presented by Lorenzo Tiacchi [8]. A parametric simulator, developed under the event/object oriented paradigm, has been embedded in a genetic algorithm for the evaluation of the objective function, which contains the simulated throughput. Banu Sungur and Yasemin Yavuz solved a new assembly line balancing problem by formulating it with an integer linear programming model, where the tasks differ according to the qualification requirements and the qualification levels of the employees were hierarchically ordered [9]. Assembly line worker assignment and balancing problem type II (ALWABP-2), two-phase variable neighborhood scanning (VNS) algorithm is proposed to solve by Olcay Polat, Can B. Kalayci, Özcan Mutlu and Surendra M. Gupta [10]. Biao Yuan, Chaoyong Zhang, Xinyu Shao, Zhibin Jiang proposed a hybrid honeybee mating optimization algorithm to minimize the number of mapped stations and total number of stations for a given cycle time in mixed-model two-sides assembly line balancing [12]. Abdolreza Roshani and Davide Giglio developed a meta-heuristic approach based on a simulated annealing algorithm to solve such an assembly line balancing problem called multi-man assembly line balancing problems type II [13]. Neda Manavizadeh, Masoud Rabbani and Farzad Radmehr addressed both balancing and sequencing problems simultaneously to achieve a combination of balancing tasks and sequencing models in mixed model production and proposed a new heuristic algorithm [14]. Abdolreza Roshani and Davide Giglio presented a cost- oriented objective function for multi-manned assembly line balancing problem [15]. A mixed-integer mathematical programming model is proposed to solve the problem optimally. Öncü Hazır and Alexandre Dolgui proposed decomposition algorithm for balancing U-type assembly lines under uncertainty [16].

The researches conducted in 2016 are as follows.

Deming Lei and Xiuping Guo proposed Variable Neighborhood Search to solve Two-sides Assembly Line Balancing Problem-II [17]. Zixiang Li, Qiuhua Tang and LiPing Zhang developed two new decoding schemes with reduced search space to balance the workload within a mated-station and reduce sequence-depended idle time.[18] A.S. Vishnu Raj, Jeeno Mathew, Peter Jose and Gishnu Sivan proposed a method to solve the

mixed model assembly line balancing problem (MALBP) with parallel workstations [19]. Dashuang Li, Chaoyong Zhang, Guangdong Tian, Xinyu Shao, Zhiwu Li presented a novel multi objective programming model for balancing a mixed-model two-sided assembly line subject to multiple constraints, in which, additional constraints including zoning, synchronous, and positional constraints are considered besides the traditional constraints, e.g., the precedence constraint [20]. A multi objective hybrid imperialist competitive algorithm is proposed to solve this problem. Zixiang Li, Qiuhua Tang, and Liping Zhang suggested an integer programming model for solving the TALBP-II with assignment restrictions optimally and utilized a simple and effective iterated greedy algorithm to address large-size problems [21]. Ullah Saif, Zailin Guan, Li Zhang, Jahanzeb Mirza and Yue Lei considered minimization of cycle time simultaneously with the maximization of the probability of stations that the flow time will not exceed the cycle time [22]. A hybrid Pareto artificial bee colony algorithm is proposed to solve the multi-objective problem by them. Zixiang Li, Mukund Nilakantan Janardhanan, Qiuhua Tang, Peter Nielsen addressed the two-sided robotic assembly line balancing problem with the objective of minimizing the cycle time [23]. A mixed-integer programming model was proposed for small-sized problems and co-evolutionary particle swarm optimization algorithm was proposed for big size problems. Pınar Tapkan, Lale Özbakır, Adil Baykasoğlu proposed Bees Algorithm and Artificial Bee Colony algorithm for parallel two-sided assembly line balancing problem with walking times [24]. Caijun Yang, Jie Gao used branch, bound and remember algorithm to solve flexible mixed-model assembly line problem with adjacent workforce cross-training policy [25]. Sebnem Demirkol Akyol and Adil Baykasoğlu proposed a constructive heuristic approach for the assembly line worker assignment and balancing problem (ALWABP) [26]. Yu-guang Zhong and Bo Ai proposed an ant colony optimization algorithm called modified ant colony optimization algorithm for multi-objective single-model assembly line balancing problem (SALBP) [27]. A multi-objective evolutionary algorithm was presented for the solution of the bi-criteria assembly line worker assignment and balancing problem (ALWABP) by P. Th. Zacharia and Andreas C.Nearchou. Qiuhua Tang, Zixiang Li and Liping Zhang used a discrete artificial bee colony algorithm by blending idle time reduction techniques for the two-sided assembly line balancing problem of type-II (TALBP-II)[28].

The researches conducted in 2017 are as follows:

Artificial Bee Colony Algorithm and a Genetic Algorithm are proposed to solve U-shaped assembly line worker assignment and balancing problem by Mehmet Kursat Oksuz, Kadir Buyukozkan, Sule Itir Satoglu [29]. Zixiang, Li Qiuhua Tang and LiPing Zhang developed two new decoding schemes with reduced search space a to balance the workload within a mated-station and reduce sequence-depended idle time [30]. They applied and evaluated a simple and effective iterated greedy algorithm. Mayron César, O. Moreira, Rafael Pastor, Alysson M. Costa and Cristóbal Miralles proposed algorithms to solve an assembly line balancing problem, which aims to combine even an employee who is sourced by situations encountered in sheltered work centers for disabled people and traditional assembly lines that require a large number of people with disabilities [31]. Abdolreza Roshani and Davide Giglio addressed the multi-manned assembly line balancing problem with the objective of minimizing the cycle time and proposed a mixed-integer mathematical programming formulation is for the considered problem [32]. Zixiang Lia, Ibrahim Kucukkoc and Qiuhua Tanga applied a new mixed-integer linear programming model for U-type assembly line balancing problem to minimize the number of opened workstations [33]. Davide Giglio, Massimo Paolucci, Abdolreza Roshani, and Flavio Tonelli addressed the balancing problem of multi-manned assembly lines with skilled workers and proposed a new mixed integer programming formulation is presented to solve this problem optimally with the objective of minimizing the total operating cost of the assembly line [34]. Lorenzo Tiacci proposed a genetic algorithm approach to balance asynchronous mixed-model U-shaped lines with stochastic task times [35]. Zixiang Li, Mukund Nilakantan Janardhanan, Qiuhua Tang and Peter Nielsen considered simultaneously balance and sequence problems robotic mixed-model assembly lines (RMALB/S), which involves three sub-problems: task assignment, model sequencing and robot allocation [36]. They proposed mathematical model and meta heuristic model to solve these problems. Celso Gustavo Stall Sikora, Thiago Cantos Lopes, Daniel Schibelbain, Leandro Magatão applied an integer-based formulation for the simple assembly line balancing problem with multiple identical tasks [37]. Mads

Kammer Christensen, Mukund Nilakantan Janardhanan and Peter Nielsen proposed a heuristic model to solve type II balance a robotic assembly line with multiple models problem (MuRALB) [38]. To solve stochastic two-sided assembly line balancing problem with multiple constraints, a hybrid teaching-learning-based optimization approach which combines both a novel teaching-learning-based optimization algorithm for global search and a variable neighborhood search with seven neighborhood operators for local search were proposed by Tang Qihua, Li Zixiang, Zhang LiPing, Zhang Chaoyong [39]. Thiago Cantos Lopes, Celso Gustavo Stall Sikora, Rafael Gobbi Molina, Leandro Magatão, Daniel Schibelbain, Luiz Carlos de Abreu Rodrigues considered balancing problem of robotic spot-welding manufacturing lines and modeled based on a real-world car factory on the outskirts of Curitiba, Brazil [40]. The model was developed with Mixed Integer Linear Programming (MILP) techniques. Maryam Salehi, Hamid Reza Maleki and Sadegh Niroomand studied a new multi-objective assembly line balancing problem [41]. The problem also is formulated in a fuzzy environment. To solve such problem, a new hybrid fuzzy interactive approach is proposed. Murat Arıkan developed a tabu search algorithm to solve the bi-objective simple assembly line balancing problem of type-2 [42]. Yin-YannChen developed a procedure of building feasible balancing solutions in automotive industry and proposed a hybrid approach based on Simulated annealing algorithm[43].

The researches conducted in 2018 are as follows:

Meral Azizoglu and Sadullah İmat considered a simple assembly line balancing problem with fixed number of workstations and prespecified cycle time and presented a mathematical model and a branch and bound procedure [44]. Parames Chutima and Krit Prasert presented a algorithm which modifies the coincident algorithm, namely the adaptive extended coincident algorithm, to solve a multi-manned assembly line balancing problem (MALBP) [45]. Fantahun M. Defersha and Fatemeh Mohebalizadehgashti proposed a mixed integer linear programming mathematical model to minimize the length and number of workstations, the costs and task copies of workstations for a continuously moving conveyor [46]. Minghai Yuan, Hongyan Yu, Jinting Huang and Aimin Ji investigated the reconfigurable assembly line balancing

problem (ALBP) in a cloud manufacturing environment based on the actual production process due to react to the increasing imbalance of assembly line due to the high uncertainty of assembly resources [47]. Mukund Nilakantan Janardhanan, Zixiang Li and Peter Nielsen researched studies worker assignment and line balancing in two-sided assembly lines with an objective of minimizing the cycle time [48]. A mixed-integer programming model is developed, to solve the small-size problems and an improved migrating birds optimization algorithm is employed to deal with the large-size problems. Zikai Zhang, Qiuhua Tang, Zixiang Li and Liping Zhang considered U-shaped robotic assembly line balancing problems [49]. First developed a nonlinear multi-objective mixed-integer programming model, and then Pareto artificial bee colony algorithm is extended to tackle complex problem. Nurhanani Abu Bakar, Mohammad Fadzli Ramli, Mohd Zakimi Zakaria, Tan Chan Sin and Hafiz Masran dealt with a case study of SALBP in power transformer manufacturing and proposed four heuristic methods in order to provide solutions for minimizing number of workstations and improve the bottlenecks simultaneously [50]. Zixiang Li, Mukund Nilakantan Janardhanan, Peter Nielsen and Qiuhua Tang developed mathematical models and simulated annealing algorithms to solve the robotic assembly line balancing problem (RALBP-II) to minimize the cycle time [51]. Zixiang Li, Mukund Nilakantan Janardhana, Amira S. Ashour and Nilanjan Dey developed two new mixed-integer programming linear models to minimize the cycle time when the number of workstations is fixed and utilized, and improved the migrating birds optimization algorithm for solving the large-sized problems for the first time in robotic U-shaped assembly line balancing problem [52]. Leonardo Borba, Marcus Ritt Cristóbal Miralles proposed a lower bound, exact and heuristic algorithms for the Robotic Assembly Line Balancing Problem (RALBP) [53]. Abhishek Sahu and S. K. Pradhan presented the comparative analysis of mixed model assembly line balancing methods and tried to find out the best possible solution in order to minimize no of workstation, ideal time, bottleneck and increase the efficiency of plant [54]. Zixiang Lia, Ibrahim Kucukkoc and Zikai Zhang developed a branch, bound and remember algorithm to address U-shaped assembly line balancing problem [55]. Honghao Zhang, Chaoyong Zhang, Yong Peng, Danqi Wang, Guangdong Tian, Xu Liu and Yuexiang Peng presented a hybrid evolutionary algorithm, namely, modified ant colony optimization inspired by the

process of simulated annealing, to reduce the possibility of being trapped in a local optimum for the balancing problem of stochastic large-scale U-type assembly line[56]. Ibrahim Kucukkoc, Zixiang Li, Aslan D.Karaoglan and David Z.Zhang introduced and mathematically formulates a mixed-model, two-sided assembly line balancing problem considering underground station [57]. José Soeiro Ferreira, Rui Rebelo and Parisa Sadeghi devised an approximate method, based on the Variable Neighborhood Descent metaheuristic and integrating an adaptation of the Ranked Positional Weighted method for Mixed-model Assembly Line Balancing Problems (MALBP) [58]. Masood Fathi, Dalila Benedita Machado Martins Fontes, Matias Urenda Moris, Morteza Ghobakhloo investigated the efficiency of the most commonly used performance measures for minimizing the number of workstations in approaches addressing simple assembly line balancing problem (SALBP) for both straight and U-shaped line, then evaluated 20 constructive heuristics to find solutions to the SALBP-1 [59]. Jiage Huo, Zhengxu Wang, Felix T. S. Chan, Carman K. M. Lee, and Jan Ola Strandhagen used a hybrid approach which executes ant colony algorithm in combination with beam search to solve the Simple Assembly Line Balancing Problem [60]. Hossein Babazadeh, M. H. Alavidooost, M. H. Fazel Zarandi and S. T. Sayyari developed a new fuzzy mixed-integer linear programming model for SALBP and SULBP [61]. Mukund Nilakantan Janardhanan, Zixiang Li, Peter Nielsen, Qiuhua Tang presented three variants of artificial bee colony algorithm to solve worker assignment and line balancing in two-sided assembly lines [62]. An immune genetic algorithm was proposed to solve the SALBP-1 for the first time that aims to minimize the number of workstations and workstation load for a given cycle time of assembly line by Han-ye Zhang [63]. To solve the balancing robotic weld assembly lines problem, a low bound of cycle time of workstations was built, and on account of the non-deterministic polynomial-time (NP)-hard nature of ALB problem (ALBP), a genetic algorithm with the mechanism of simulated annealing, as well as self-adaption procedure, was proposed to overcome the inferior capability of genetic algorithm in aspect of local search by Binghai Zhou and Qiong Wu.[64] Ibrahim Kucukkoc addressed to mixed-model two-sided lines incorporating incompatible task groups and proposed a new method for minimizing two conflicting objectives, namely cycle time and the number of workstations, to maximize line efficiency [65]. Jordi Pereira

and Marcus Ritt and Óscar C. Vásquez proposed a hybrid metaheuristic to solve the problem, which denoted as the cost-oriented robotic assembly line balancing problem [66]. M. Lalaoui and A. El Afia proposed an adaptive generalized simulated annealing using fuzzy inference system to solve simple assembly line balancing problem of type I (SALBP-I) [67]. Heydar Ali Mardani-Fard, Abdollah Hadi-Vencheh, Ali Mahmoodirad and Sadegh Niroomand proposed an effective hybrid goal programming approach for multi-objective straight assembly line balancing problem with stochastic parameters [68].

The researches conducted in 2019 are as follows:

Markus Pröpster, Lothar März, Gunther Reinhart and Carsten Intra presented a simulation tool which simulates these aspects of worker flexibility according to the produced variants and their sequence [69]. Hassan Gharoun, Mahdi Hamid, Seyed Hossein Iranmanesh and Reza Yazdanparast used an intelligent algorithm for Performance improvement of two-sided assembly line balancing problem Considering Learning Effect and Allocation of Multi-Skilled Operators [70]. Dilruba Telli, Yalçın Berberoğlu, Gizem Karadeniz, Doğangün Yılmaz Uzun, Elif Baykal and Damla Yüksel considered modify the 5th lean assembly line in order to include a new product model in a combi boiler producer company [71]. The solutions are developed by various techniques and approaches such as Bubble diagram, Value Stream Mapping, Lean Production, Line Balancing. Omar Samak, Mehmet Eren Nalici and Ramazan Duran suggested two approaches addressing line balancing and energy wastage are studied and obtained positive results with simulation [72]. For simple assembly line balancing problem, Shady Magdy Anwer, Ahmed Mahmoud Ali and Mohamed Ahmed Awad proposed Enhanced Genetic Algorithm using precedence preservative crossover and scramble mutation techniques, aiming to minimize two objectives; number of stations as a primary objective and smoothing index as a secondary objective [73]. Fahimeh Tanhaie, Masoud Rabbani and Neda Manavizadeh studied a mixed-model assembly line balancing problem is applied in a make-to-order environment, and to solve this problem, proposed a non-dominated sorting genetic algorithm [74]. Srisawat Supsomboon aimed to reduce bottlenecks and increase productivity in wax pattern and casting processes

using line balancing [75]. He used simulation for solving problem. Anikó Bartos and Botond Bertok presented process graphs or P-graphs for leading in an easy to follow visual definition of the potential task to employee allocations, as well as the options to generate a mathematical programming model algorithmically [76]. Yuchen Li, Yao Fu, Xiaowen Tang and Xiaofeng Hu utilized uncertainty theory to model task time uncertainties and introduced the belief reliability measure to the assembly line production for the first time [77]. Michela Dalle Mura and Gino Dini proposed a multi-objective genetic algorithm for solving the assembly line balancing problem considering ergonomics based on energy expenditure [78]. Mingshun Yang, Li Ba, Erbao Xu, Yan Li, Yong Liu and Xinqin Gao studied a multi-constraint two sides assembly line balancing problem (TALBP) that involves positional constraints, zoning constraints and synchronism constraints to make TALBP more in line with real production, and proposed an improved imperialist competitive algorithm [79]. Abolfazl Jafari Asl, Maghsud Solimanpur and Ravi Shankar proposed a multi-objective mixed-integer linear programming model for balancing multi-objective multi-model assembly lines [80]. Wucheng Yang and Wenming Cheng considered problem of balancing mixed-model two-sided assembly lines with setup [81]. Firstly, they developed a mixed-integer programming (MIP) mathematical model and then, used effective variable neighborhood search algorithm for second option. Zixiang Li, Mukund Nilakantan Janardhanan, Qiuhu Tang and S.G. Ponnambalam developed new mixed integer linear programming model with the objective of optimizing the cycle time on the robotic two-sided assembly line with consideration of sequence-dependent setup times and robot setup times [82]. K Dziki and D Krenczyk presented several metaheuristic methods that can be used to solve line balancing and tasks assignment problems and show the advantage of simulated annealing over other methods in solving the problem [83]. Zikai Zhang, Qiuhua Tang and Liping Zhang proposed mathematical model and grey wolf optimization for low-carbon and low-noise U-shaped robotic assembly line balancing problem [84]. F.H. Ho, M. Al-Haqeem Chee S. Abu and Y. L. Woo developed a balanced synchronized matrix for contribute to enable better understanding of line balancing in powder coating process [85]. Reza Ramezani and Sadjad Khalesi considered the problem of integration of multi-product Supply Chain Network designing and Assembly

Line Balancing and proposed a mixed integer nonlinear programming model to formulate the studied problem [86]. Mohd Fadzil Faisae Ab. Rashid, Ahmad Nasser Mohd Rose, Nik Mohd Zuki Nik Mohamed and Fadhlur Rahman Mohd Romlay propose an improved Moth Flame Optimization algorithm to optimize the cost-oriented two-sided assembly line balancing [87]. Yahui Zhang, Xiaofeng Hu and Chuanxun Wu proposed a mathematical model and an improved imperial competition algorithm to solve the multi-objective two-sided assembly line rebalancing problem with space and resource restrictions [88]. Thiago Cantos Lopes, Giuliano Vidal Pastre, Adalberto Sato Michels and Leandro Magatão proposed flexible station frontiers for multi-manned lines and showed that such innovation allows significantly shorter line lengths [89]. A new Mixed Integer Linear Programming model and a novel model-based heuristic procedure are presented to describe and optimize lines. Haile Sime, Prabir Jana and Deepak Panghal used simulation technique for line balancing in apparel industries [90]. Carlos Alexandre X. Silva, Les Foulds and Humberto Longo proposed a mathematical model of the Task Division Assembly Line Balancing Problem, an exact solution procedure for it present promising computational results for the adaptation of some classical SALBP instances from the research literature [91]. Nima Mirzaei, Ali Mahmoodirad and Sadegh Niroomand proposed a credibility-based fuzzy modeling approach for an uncertain multi-objective assembly line balancing problem [92]. Masood Fathi, Amir Nourmohammadi, Amos H. C. Ng and Anna Syberfeldt aimed to bridge the gap between theory and practice by addressing a real-world assembly line balancing problem (ALBP) where task times are stochastic and there are zoning constraints in addition to the commonly known ALBP constraints [93]. A mixed integer programming (MIP) model is proposed. Thiago Cantos Lopes, Celso Gustavo Stall Sikora, Adalberto Sato Michels and Leandro Magatão presented a new mixed-integer linear programming model to describe and optimize the simultaneous balancing, cyclical sequencing and buffer allocation asynchronous assembly line problem [94]. Binghai Zhou and Qiong Wu developed an improved immune clonal selection algorithm for bi-objective robotic assemble line balancing problems considering time and space constraints [95]. Masood Fathi, Amir Nourmohammadi, Amos H.C. Ng, Anna Syberfeldt and Hamidreza Eskandari proposed an improved genetic algorithm with variable neighborhood search to solve the assembly

line balancing problem [96]. Gokhan Secme and Lale Özbakır adapted and applied a simulated annealing-based hyper-heuristic to the balancing problem of oven manufacturing process with assignments and operational restrictions with multiple objectives [97]. Amir Nourmohammadi, Masood Fathi, Mostafa Zandieh and Morteza Ghobakhloo used a water-flow like algorithm for solving U-Shaped assembly line balancing problems [98]. M. Lalaoui and A. El Afia used a reactive generalized simulated annealing (GSA) using type-2 fuzzy controller in charge of adapting dynamically the generalized simulated annealing parameters during the search for to respond to demand fluctuation on Mixed-model assembly lines [99]. Öncü Hazır, Maher Agi and Jérémy Guérin developed a problem specific heuristic algorithm to solve smoothing simple assembly line with the large instances effectively [100]. Daniel Neumann and Jan Keidel built the dependency map and then integrated within an existing assembly line planning process [101]. Ibrahim Kucukkoc, Kadir Buyukozkan, Sule Itir Satoglu and David Zhang proposed a mathematical model and artificial bee colony algorithm for the lexicographic bottleneck mixed-model assembly line balancing problem [102]. Adalberto Sato Michels, Thiago Cantos Lopes, Celso Gustavo Stall Sikora and Leandro Magatão studied the multi-manned assembly line balancing problem and presented a new Mixed-Integer Linear Programming model with strong symmetry break constraints and decomposes the original problem into a new Benders' decomposition algorithm to solve large instances optimally [103].

The researches conducted in 2020 are as follows:

Yuchen Li, Ibrahim Kucukkoc and Xiaowen Tang utilized the uncertainty theory to model uncertain task times, considered incompatible task sets constraints and solved by using a simulated annealing algorithm [104]. Ashish Yadav and Sunil Agrawal applied spider monkey optimization algorithm to solve benchmark problems of two-sided assembly line balancing problem to minimize the number of mated stations and idle time [105]. Masoud Rabbani, Seyedeh Zeinab Beladian Behbahan and Hamed Farrokhi-Asl used an Augmented Multi-Objective particle swarm optimization to solve mixed-model four-sided assembly line balancing problem for human-robot collaboration [106]. Nurhanani

Abu Bakar, Mohammad Fadzli Ramli, Mohd Zakimi Zakaria, Tan Chan Sin and Muhamad Hafiz used four heuristic methods for minimizing number of workstations and improve the bottleneck problems at the same time [107]. Ganokgarn Jirasirilerd, Rapeepan Pitakaso, Kanchana Sethanan, Sasitorn Kaewman, Worapot Sirirak and Monika Kosacka-Olejniak presented a variable neighborhood strategy adaptive search method in a case study of the garment industry considering the number and types of machines used in each workstation in a simple assembly line balancing problem type 2 [108]. Ashish Yadav and Sunil Agrawal proposed a mixed integer programming formulation for multi-manned assembly line balancing problem which a group of workers simultaneously performs different tasks on a workstation [109]. Ashish Yadav, Pawan Verma & Sunil Agrawal proposed objective function to maximize the workload at each station such that the number of stations is minimized in mixed model two-sided assembly line [110]. Ahmad Tavakoli used hybrid Tabu-SA algorithm for multi-criteria optimization of multi product assembly line [111]. Susanto Yaphiar, Cahyadi Nugraha and Anas Maruf presented a new mathematical model for mixed model assembly line balancing that includes human-robot collaboration [112]. Mohammad Huskhazrin Kharuddin, Mohammad Fadzli Ramli and Hafiz Masran used heuristic procedure and simulation to solve the line balancing problem [113]. Zixiang Lia, Ibrahim Kucukkoc and Zikai Zhang proposed a new branch, bound and remember (BBR) algorithm to minimize the number of mated-stations in two-sided assembly lines [114]. Kai Meng, Qiuhua Tang, Zikai Zhang and Xinbo Qian proposed an improved lexicographical whale optimization algorithm for the type-II assembly line balancing problem considering preventive maintenance scenarios [115]. Y. Lahrichi, Nathalie Grangeon, Laurent Deroussi and S. Norre proposed a polynomial exact algorithm for the reconfigurable transfer line balancing problem [116]. Öncü Hazır, Maher A.N. Agi and Jérémy Guérin developed an efficient heuristic to solve the workload smoothing problem in simple assembly lines by investigating and integrating the problem-specific characteristics [117]. Samah A. Aufy and Alla Eldin H. Kassam proposed a consecutive heuristic algorithm for balancing a mixed - model assembly line type II using a worker – task assigned to workstation heuristic model developed for straight and U-shaped layouts [118].

1.2 Purpose of Thesis

Assembly lines are an important part of manufacturing systems that affect overall cost and efficiency. The productivity of the human resource in assembly lines directly affects the efficiency of the lines in practice.

This paper addresses Simple Assembly Line Balancing Problem (SALBP) Type 2 where the number of workstation lines is known. The main objective is balancing the cycle time and physical workload of the stations simultaneously. Parallel task assignment and multi-manned workstations method from the literature is proposed to increase line efficiency. Result: A non-linear mixed integer programming model is developed to solve the problem and after that model is tested on 2 different scenarios to illustrate the methodology.

1.3 Hypothesis

Difficulties occur when trying to implement a real world assembly line balancing problem with the goal of improving line efficiency, to a model, provided by a study. If all the parameters and assumptions are chosen carefully with Parallel task assignment and Multi-manned workstations approaches, a model to solve a real world problem can be created.

1.4 Content of Thesis

Assembly line balancing problems relate to the proper allocation of works to workstations, considering the priority relationships within the line and optimizing a specific objective function. In other words, in assembly line balancing problems, it is aimed to efficiently group facilities or employees to achieve optimum balance between the capacities and flows of assembly processes. This problem is a problem that occurs on the production line, which is classified as one of the difficult optimization problems.

In this context, assembly line balancing problems are one of the most important points of production for businesses.

In this study, we discussed the Simple Assembly Line Balancing Problem (SALBP) Type 2. In order to solve the bottleneck problem in the assembly line, approaches of parallel task assignment and multi-manned workstations are added to the model. After coding this model to obtain results via python using linear programming method, deficiency of the model revealed itself. Since, optimum results could not be achieved, we adapted the non-linear programming model to the problem by adding approaches of parallel task assignment and multi-manned stations. Furthermore, Model was evaluated with sensitivity analysis.

In the first part of this study, the literature review with the subject, in the second part, theoretical information about assembly lines and assembly balancing problems, in the third part, the details of our model that was worked on, in the fourth part, the result and suggestions of this study are included.

2. GENERAL INFORMATION

CHAPTER II

In this section, assembly lines, assembly line balancing and value flow mapping are the subjects that are the basis of the application of the thesis.

2.1 Definition and Classification of Assembly Lines

The assembly line is the industrial arrangement of machinery, equipment and workers for the continuous flow of workpieces in mass production operations. The design for an assembly line is determined by analyzing each product component as well as the steps required to produce the final product. [119] The entire movement of the material is simplified without cross flow, backtracking or repetitive procedure. Job assignments, number of machines and production rates are programmed to be compatible with all operations on the line, in other words, an assembly line is a production process that separates the production of a good into steps completed in a predefined order.

Assembly lines are the most widely used method of mass production of products. They can lower labor costs because unskilled workers can be easily trained to perform certain tasks. Instead of hiring a skilled master to assemble an entire furniture or vehicle engine, companies hire a worker to just add a stool or bolt leg to a machine. [120] Another reason that many products are still manually assembled is that many component parts cannot be easily transported by machines. Expensive and somewhat inelastic automatic assembly machines are economical only if they produce high levels of output. However, with the development of versatile machines, the use of industrial robots in assembly lines is increasing. Increased use of industrial robots increases the efficiency of fully automated assembly operations.[119]

Assembly lines are classified in many ways in the literature. If these literature studies are considered in the most inclusive way, assembly lines can be classified under six main titles as shown in the Figure 1.

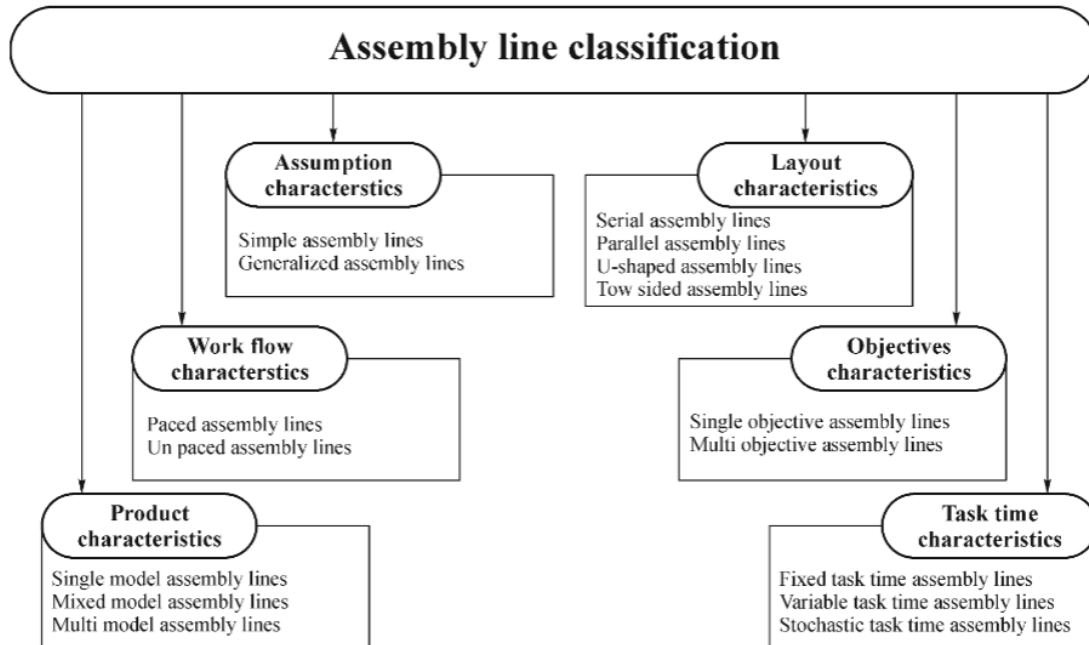


Figure 1: Assembly Line Classification

2.1.1 According to Assumption Characteristics

A well-known classification method separates assembly line problems into two classes [121]. It is called the simple assembly line balancing problem (SALBP) and the first-class generalized assembly line balancing problem (GALBP) called the second. For example, it is assumed that the stations are aligned in a straight line along the conveyor and a uniform product is mounted on it, the task duration, product demand and other input parameters are known and deterministic in advance. Problems that try to get more realistic results by eliminating one or more of these assumptions are in the generalized assembly line balancing problem (GALBP) category.

2.1.2 According to Work Flow Characteristics

Assembly lines can be categorized into two types based on the workflow features. These include paced assembly lines and un-paced assembly lines. In paced assembly lines, all stations are assumed to have similar cycle time (CT) and upon the end of each cycle time, each station moves the products to the next station after finishing its tasks. In un-paced asynchronous assembly lines, each station may have different cycle time which is equal to the workload of that station. In these systems when parts are finished on a station, there is possibility that the next station may have not finished its work. [122]

2.1.3 According to Product Characteristics

Assembly lines are divided into three different types based on their product characteristics: [122]

In single model assembly lines, a large quantity of only one kind of products assembled. The task time and precedence constraints are represented by a single precedence graph in this case.

In a mixed model assembly lines, different models which are slightly different from each other but have the same parent product are assembled. Different models can have different process constraint each defining its model task durations and precedence relations.

In multi model assembly lines, different models are assembled in batch form. These products are not mixed with each other during production but are produced in batches.

2.1.4 According to Layout Characteristics

Based on layout, assembly lines can be categorized into various kinds such as serial lines, parallel lines, U-shaped lines, and two-sided assembly lines. [122]

In serial assembly lines, stations are arranged in a serial manner along the sides of the conveyer. Parts are moved into the assembly line and moved to first station. After finishing their processing on the first station they are moved down the line one by one from the current to the next and leave the line after passing through last station.

Parallel assembly lines are used to divide the workload among stations [123]. For example, if the cycle time of the line is more than desired, then the workload of the station which has maximum workload can be divided by paralleling this station. It can help to reduce the cycle time of the line.

Milten-burg and Wijngaard [124] introduced U-shaped assembly lines in which stations arranged in U-shaped layout. One of the advantages of U-shaped assembly line arrangement is that, some stations can be revisited for some tasks of assembly. Furthermore, some stations which are idle can be utilized to perform some more tasks and it can help to improve station utilization.

Another type of assembly line based on layout designs is the two-sided assembly lines. These lines are designed to assemble heavy and large sized products like automobiles, trucks, busses and large construction machinery. Two-sided assembly lines assembled parts from the both sides of assembly lines [125,126,127] and more than one worker or machine are working together on a part simultaneously on a station.

2.1.5 According to Objectives Characteristics

Assembly lines can be classified into two categories according to the objective functions used for their optimization. Single objective optimization problems of assembly line are very common and several research articles consider different kinds objectives such as minimization of cycle time or minimization of number of workstations to optimize. In multiple objective assembly line, on the other hand, try to obtain more than one objective at the same time.

2.1.6 According to Task Time Characteristics

Task time characteristics divide assembly lines into three groups i.e., fixed task time, varying task time and stochastic task times. When the expected variation in task time is small task time is considered as deterministic. Indeterministic environment task times is assumed as a fixed variable and it is assumed that, it is not changed during assembly operations. This assumption of task time makes the assembly lines simple to solve.

However, in real assembly lines, task time varies. Task time changes due to several reasons, for example, when a task is given to a manual labor, he can perform the task in some duration of time and after getting an experience in doing that task, he may take less time to perform the same task. This systematic reduction of task time is possible due to learning effect [128,129]. In this kind of problems, task time is not fixed and is considered as a variable which depends on the learning rate of the labors.

In stochastic assembly line balancing problem task time is considered as unknown variable as stochastic variable, [130,131] fuzzy variable [132,133], and independent normal distributed variable [134] or some other distributions [135,136]

2.2 Assembly Line Balancing Problem

Line Balancing generally means balancing the production line or any assembly line. The main purpose in line balancing is to minimize the running time of the system by distributing the work times equally to the stations. In other words, it aims to group facilities or employees efficiently to achieve the optimum or most efficient balance between the capacities and flows of production or assembly processes.

Assembly line balancing problems are related to the proper allocation of works to the workstations in a balanced manner, considering the priority relationships within the line and optimizing a specific objective function. This problem is a problem that is classified as one of the difficult optimization problems and mainly occurs on a continuous production line. [122] Installation of the assembly line is a long-term decision and high cost. Therefore, it is necessary to design the assembly line in the enterprises and balance the workload at the workstations.

2.2.1 Classification of Assembly Line Balancing Problem

Assembly line balancing problems can be classified in different ways according to their characteristics. It is for classifications based on objective functions and problem structure. Assembly line balancing problems are diversified in terms of developing technology and expectations. Accordingly, classifications have been expanded in the

literature and different approaches have been presented. Below are schemes of some of these classifications.

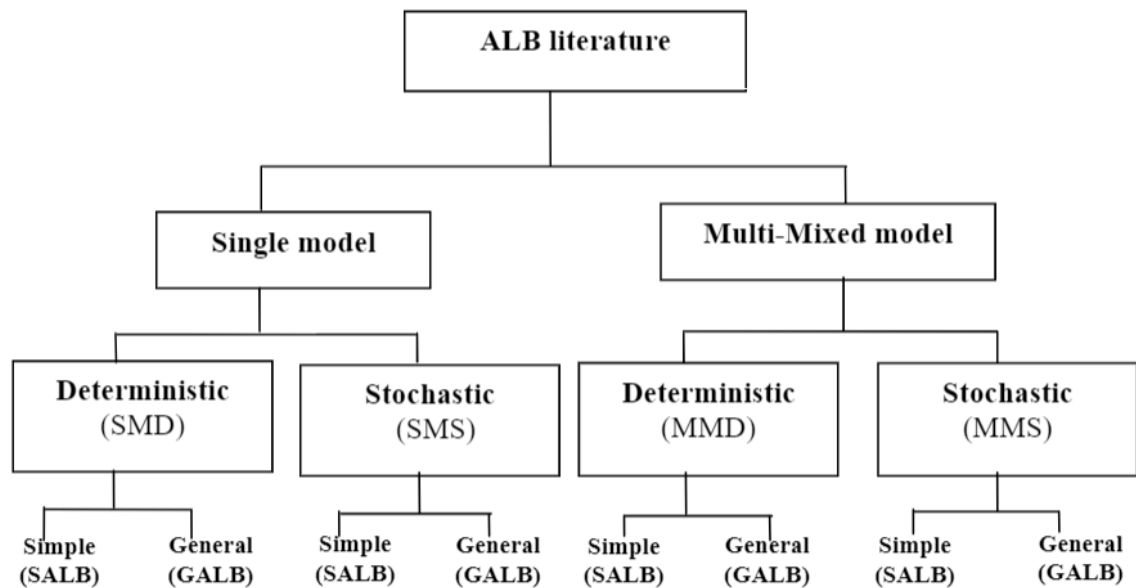


Figure: 2 Classification of Assembly Line Balancing According to Ghosh and Gagnon, 1989

Ghosh and Gagnon (1989) classified the ALBP into four categories; as shown in Figure 2: Single Model Deterministic (SMD), Single Model Stochastic (SMS), Multi/Mixed Model Deterministic (MMD), Multi/Mixed Model stochastic (MMS) [121].

The simplest form of assembly line balancing problems is a single model deterministic task time line balancing problem and is known as a simple line balancing problem. The main assumptions of the problem are: [138]

- One kind of product on the assembly line is produced in large quantities.
- Term of office is deterministic and known.
- Priority relationships between tasks are known.
- An entire task must be performed at one station.
- If a task has a predecessor, all its predecessors must have been completed before that task can begin.

- The single-model line balancing problem has three types of constraints, namely assignment constraints, cycle time constraints, and priority relationship constraints.

The production style in which different models of different products or of the same product are produced or assembled on the same line is called multi-model or mixed-model production.

- In multi-model production, various models are produced in batches on the same line.
- In the mixed model production, the batch size in the line can decrease up to one.

The success of the mixed model production depends on the effective solutions of two important problems called “mixed model line balancing” and “model sorting”. The difference between the mixed-model assembly line balancing problem and the single-model assembly line balancing problem, or even because the model produced is more than one, differs between the models of the tasks, duration and priority relationships. The model sorting problem is related to determining the order in which the various models produced on the assembly line will be produced.

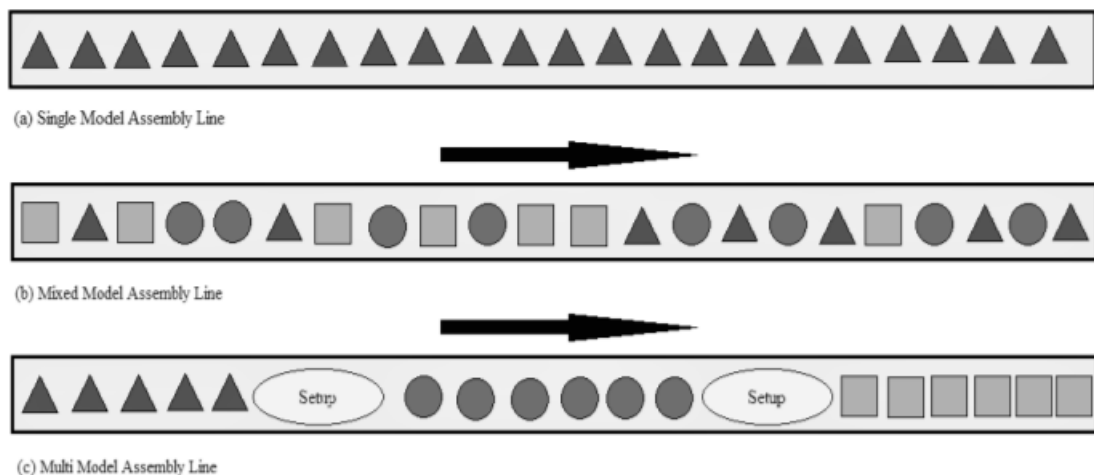


Figure: 3 Assembly Lines For Single, Mixed and Multiple Products

According to Becker and Scholl, assembly line balancing problems should basically be divided into two [139]: Simple assembly line balancing problems and general assembly line balancing problems.

SALBP: The simple assembly line balancing problem is relevant for straight single product assembly lines where only precedence constraints between tasks are considered.

- Type 1 (SALBP-1) of this problem consists of assigning tasks to workstations such that the number of stations (m) is minimized for a given production rate (fixed cycle time).
- Type 2 (SALBP-2) is to minimize cycle time (maximize the production rate) for a given number of stations.
- Type E (SALBP-E) is the most general problem version maximizing the line efficiency (E) thereby simultaneously minimizing c and m considering their interrelationship.
- Type F (SALBP-F) is a feasibility problem which is to establish whether or not a feasible line balance exists for a given combination of number of station and cycle time, [4]

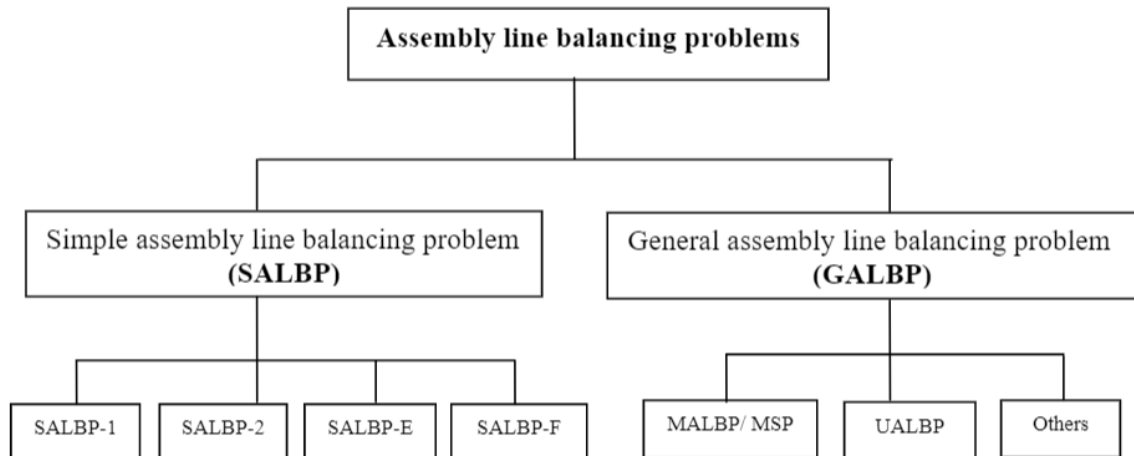


Figure: 4 Classification Assembly Line Balancing According to Scholl and Becker, 2006

GALBP: In the literature, all problem types which generalize or remove some assumptions of SALBP are called generalized assembly line balancing problems (GALBP). This class of problems (including UALBP and MALBP) is very large and

contains all problem extensions that might be relevant in practice including equipment selection, processing alternatives, assignment restrictions etc.

- MALBP and MSP: Mixed model assembly lines produce several models of a basic product in an intermixed sequence. Besides the mixed model assembly line balancing problem (MALBP), which has to assign tasks to stations considering the different task times for the different models and find a number of stations and a cycle time as well as a line balance such that a capacity- or even cost-oriented objective is optimized (cf. Scholl, 1999, Chapter 3.2.2). However, the problem is more difficult than in the single-model case, because the station times of the different models have to be smoothed for each station (horizontal balancing; cf. Merengo et al., 1999). The better this horizontal balancing works, the better solutions are possible in the connected short-term mixed model sequencing problem (MSP). MSP has to find a sequence of all model units to be produced such that inefficiencies (work overload, line stoppage, off-line repair etc.) are minimized. (e.g. Bard et al., 1992 and Scholl et al., 1998)

- UALBP : The U-line balancing problem (UALBP) considers the case of U-shaped (single product) assembly lines, where stations are arranged within a narrow U. As a consequence, worker are allowed to work on either side of the U, i.e. on early and late tasks in the production process simultaneously. Therefore, modified precedence constraints have to be observed. By analogy with SALBP, different problem types can be distinguished. (cf. Miltenburg and Wijngaard, 1994; Urban , 1998; Scholl and Klein,1999; Erel et al., 2001)

2.2.2 Assembly Line Balancing Methods

Assembly line balancing methods can be classified into three main groups as analytical methods, heuristic methods and simulation techniques [140].

2.2.2.1 Analytical Methods

Analytical methods are also known as precise methods or optimization methods. In this method, the aim is to find the most suitable solution for the problem [140]. It takes time to get solutions in analytical methods, but the result is reached analytically. In addition,

the solution time of the problem changes in direct proportion to the size of the problem. In other words, as the size of the problem grows, the solution time increases in comparison with this growth.

Some of the exact methods are as follows [140].

- Mathematical Programming
- Dynamic Programming
- Loosened Lagrange Method (Lagrangian Relaxation)
- Position Weight Technique
- Branch-Bound Algorithm
- Local Lower Limit Method

2.2.2.2 Intuitive (Heuristic) Methods

Heuristic (heuristic) methods are methods developed according to the nature of the problem. These algorithms are used without certainty in these methods. These algorithms do not always work with the same performance and do not always guarantee results or that the calculated result is the most appropriate. But to the solution; they are preferred because they are reached more quickly with less calculation under specified constraints. It can also be considered that they are relatively more valid and give better results.

Intuitive algorithms are divided into two according to the solution method. In the first of these solution methods; rules are determined before the solution of the problem. According to the determined rules, problem solving is performed. Then, assignments are made to workstations. Some of such heuristic algorithms are as follows [140]:

- Zoning Method (Killbridge-Wester)
- Position Weight Management (Helgeson-Birnie)
- Moodie-Young Method
- Ordered Position Weights Method

- Critical Path Method
- Assignment Method According to Scores Ordered from Big to Small
- Hoffmann Intuitive Method

In the other heuristic solution method; Firstly, the initial solution method is determined. Assignments are made to the stations with the rules determined according to this method. Some of such intuitive algorithms are as follows [1].

- Genetic Algorithms
- Simulation Annealing
- Taboo Search
- Particle Swarm Optimization
- Ant Colony Optimization
- Weighted Super Position Shooting
- Forest Optimization Algorithm
- Artificial Bee Colony

2.2.2.3 Simulation Techniques

It is the establishment of a computer model that mimics the system's operation over time in order to understand the mechanism of any system or evaluate different strategies related to the operation of the system. Various experiments with the established computer model are simulation techniques. It is generally used to determine intermediate stock levels. Although the most suitable solution is found with the mathematical method, the solutions found are more optimal because the solution found requires many probability calculations.

The simulation technique steps are as follows [140].

1. Identification of the problem.
2. Description of the system.
3. Converting the system to flow chart.

4. Translation of the model into a language understandable by the computer.
5. To determine whether the program running on the computer is working correctly.
6. Determining the acceptability of the results obtained from the model.
7. Conducting an experiment to check that the requested information has not been obtained.
8. Determining how each experiment will take place.
9. Experimenting the production of the requested data with the simulation model.
10. Interpretation of the produced data.
11. Putting model information into use.
12. Documenting the information about the model by recording

2.3 Value Stream Mapping

Value-stream mapping, also known as "material- and information-flow mapping"[141], is a lean-management method for analyzing the current state and designing a future state for the series of events that take a product or service from the beginning of the specific process until it reaches the customer. A value stream map is a visual tool that displays all critical steps in a specific process and quantifies easily the time and volume taken at each stage.[142] Value stream maps show the flow of both materials and information as they progress through the process.[143] It describes the current production conditions in an understandable manner. Starting from a manually recorded current state map, an ideal state map is proposed.

The purpose of value-stream mapping is to identify and remove or reduce "waste" in value streams, thereby increasing the efficiency of a given value stream. Waste removal is intended to increase productivity by creating leaner operations which in turn make waste and quality problems easier to identify. [144] In other words, VSM can be described as a classical pen-and-paper method, which maps the "value-adding, non-value-adding, and value preserving activities that are required to create a product". The

VSM method currently faces a “diminishing gradient” of effectiveness: The identification and elimination of waste and inefficiencies has become more complex, because the “low hanging fruits” have already been picked in many factories. [145]

The difference between a value stream and a value chain is that a value stream focuses only on areas of a firm that add value to a product or service, whereas a value chain refers to all the activities within a company.[146]

2.3.1 Steps of Value Mapping

Step 1: Gather Preliminary Information

Before moving ahead with VSM, be sure to gather the following information:

- A history of your product mix from the previous year's sales
- The volume of sales for each product sold from the previous year
- A list of your annual sales goals for the next year
- Any calculated assumptions from your sales and marketing strategy that will affect your sales for the next year

Step 2: Create a Product Quantity Routing Analysis

To create a product quantity routing analysis, start by listing all your customers and the products you build for each of them. Be sure to stick to the basics when it comes to your products. Don't get bogged down with the various options that can be added to your base product. List your base standard products that differ in form, fit, and function.

Step 3: Group Customers and Sort Materials

The next step is to group your customers together by the industry they represent. Sort out the number of material types by each industry and determine how many parts account for each type. The purpose of this exercise is to identify which product families you want to map.

Step 4: Sort Product Families by Build Sequence

The next step before VSM is to construct a process quantity routing analysis to identify product families according to a similar build sequence. With this information, you will know exactly what you are mapping through the flow of your operations.

Step 5: Choose One Value Stream to Begin With

It is important to know what products or piece parts comprise 80 percent of your business and which products or piece parts of that 80 percent make sense to group together as a value stream (product family). Start by value stream mapping one value stream at a time to keep things simple. You may find that although your analysis is examining 80 percent of what you build, it might be useful to group together products or piece parts from the other 20 percent with other value streams. This could arise because they follow the same build sequence and have similar form, fit, and function. Don't be so militant in following the 80-20 rule that you don't leave room for flexibility when opportunities arise that are logical to include in your analysis

Step 6: Create an Operations Flow Chart

Next, create a flow chart of all the operations in your value stream. Refer to work order routings or the bill of materials for the product value stream as needed to identify the steps and create a starting point.

Step 7: Walk the Shop Floor

When it's time to map the product family, the team needs to take a walk around the shop floor. Walk through the steps of your value stream where the work is being done. Start at the end of the process, working your way upstream to the beginning, to get a customer-supplier perspective at each step. You will be able to see any waste that's occurring and whether it is unnecessary or acceptable. Don't rationalize waste and assume it is happening only because of a new operator in training or a new product design, as you will miss identifying opportunities for improvement.

Step 8: Collect the Data

The first step is to refer to the routings you have available in your MRP system. Conduct time studies where applicable and review the results with the operators. If the operators

and your team agree with your results, then use this data to fill in the gaps in your data box for any process that is missing data. Operators can be the best source for information when the data is unavailable, and you can't conduct a time study.

Another option is to use data from similar jobs. Referring back to your process quantity routing analysis, you should be able to identify other part numbers in your value stream that are similar. Run a report on a handful of similar parts in your value stream to give you some data to examine.

Step 9: Construct the VSM

Metrics and Measurements. On a separate sheet of blank paper, begin constructing your value stream using symbols to indicate what is happening at each process.

At each step, be sure to measure the appropriate metrics as they apply to your process. This paints a good picture of the current condition at each process, despite what your MRP system says or what others might tell you.

Step 10: Summarize the Data and Get the Big Picture

When you have identified all the steps in your value stream and have filled in all the data applicable to your VSM, it's time to put it all together. Remember, the point isn't to put together an artistic portrait of the flow through your value stream, but to use your VSM to see where the value is being created and identify where wasteful activity occurs. To help you find the waste, examine the process and ask yourself, "If the customer were to walk through each step of this value stream with his checkbook in hand, would he be willing to pay for each step he sees" [147]

3. METHODOLOGY

CHAPTER III

3.1 Description of the study area

Subject of the study is one of the world's leading companies in the sector and the one of the largest home appliances manufacturer in Europe. To be the first choice for consumers worldwide drives the company and its employees. The company aspires to improve the quality of life for people with its exceptional brands, high-class products and superior solutions.

Company has plants that manufactures many different types of home appliances. The company's high quality level plant is analyzed in this study.

In the metioned plant of this company, two different types refrigerator/cooler are beeing manufactured. These types are; SD (single door) and BM (bottom mount). First one, as it can be interpreted from the name, represents refrigerators with only one door. The latter one represents refrigerstor with two or three doors.

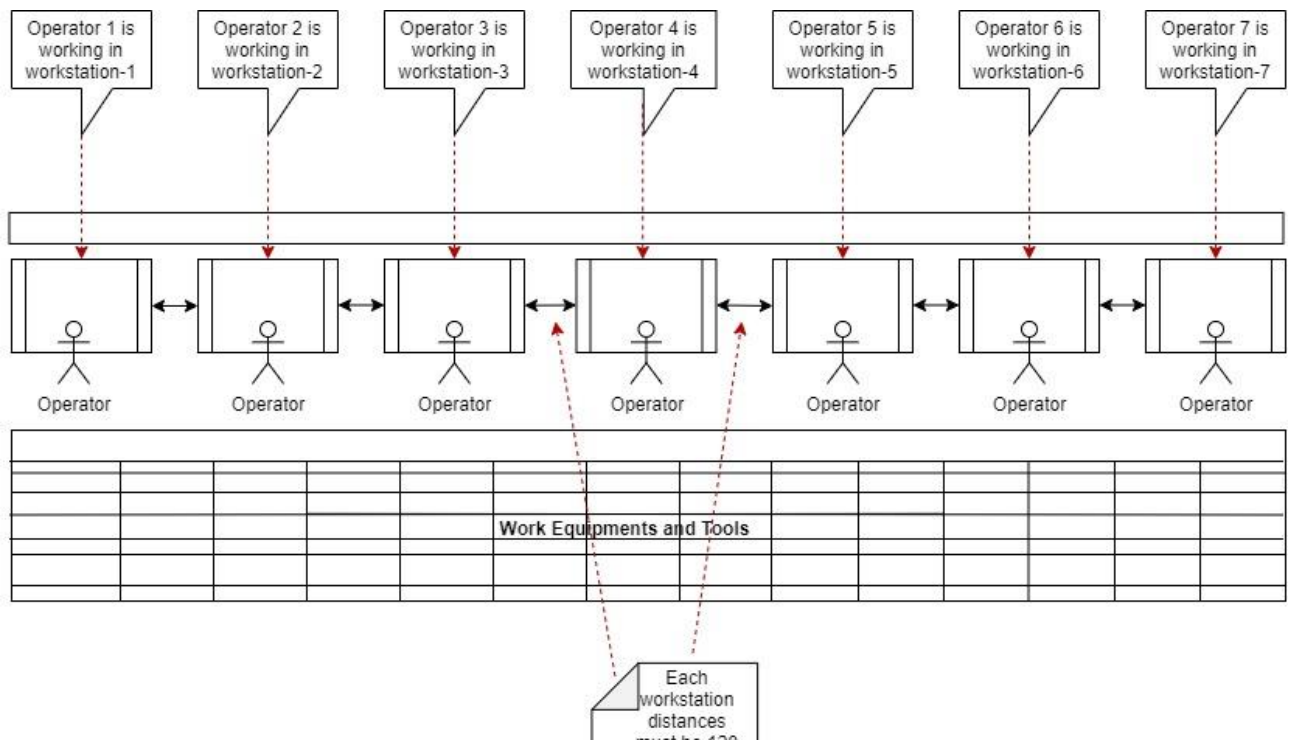


Figure 5: Representation of Work Area

These refrigerators are manufactured through various lines which mainly are pre-assembly, 180°converter, body-foaming, assembly (horizontal line, vertical line, aggregate), PMS, packaging, final control. While non-value adding processes are being done in pre-assembly, key mechanisms and processes that make refrigerator perform, like evaporator, are being done in assembly.

This study examines horizontal line which is the first line of assembly. In the horizontal line there are five work-stations. Each station has an operator working on it. The distance between each workstation must be at least 120 cm.

Actually the process is more complex, in the 3.5 Value Stream Mapping section all processes can be seen better.

3.2 Current Production Planning Process

In current situation, planning activities are being made by engineering division of plant.

Engineering division gives yearly updated work instructions that include task orders and emergency protocol to operators. Because the division is concerned with the whole plant, horizontal line is not optimized and becomes a bottleneck. This prevents plant to reach the desired cycle time in horizontal line so the plant fails to increase its production capacity.

3.3 Problem Definition

This paper is concerned with a problem of Single Model Assembly Line Balancing Problem Type-II (SALBP-II) which minimizes the cycle time for a given number of workstations. Type II problems deal with the maximization of the production capacity rate of an existing assembly line. The MALBP-II is applied when, for example, changes in the assembly process or in the product range require the line to be redesigned.

The problem addressed in this research is the mixed-model assembly line balancing problem of type-II that has a set of similar models which must be simultaneously assembled. Each model has its own set of precedence relationships, but it is possible to

combine all the relationships into one precedence diagram with various number of tasks. Each task has a processing time that may vary between models.

The problem of this study has a goal of maximizing efficiency of the line that is being analyzed by decreasing cycle time. Company has a goal of achieving below 50 seconds cycle time in all the workstations. Furthermore, getting additional operator and workstation is impossible due to 120 centimeters distancing rule between workstations that limits available space. In addition to that when an operator finishes the task on the station, operator in the next station may still be continuing his own task.

3.4 The Methodology

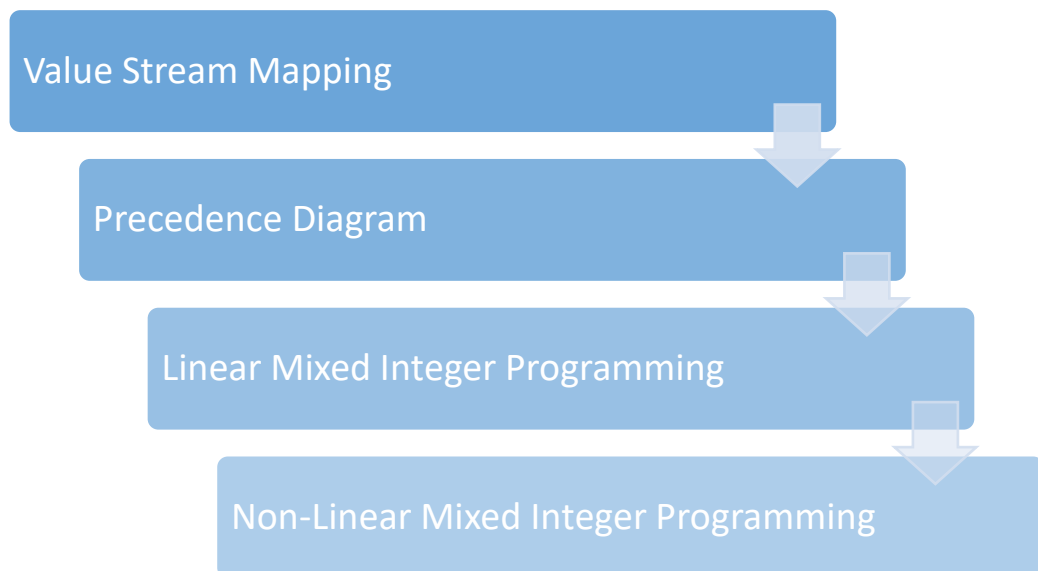


Figure 6:Methodology Steps

In this study, the following steps are followed as methodology steps.

3.4.1 The Value Stream Mapping

In the framework of this methodology, firstly the value flow mapping(VSM) technique is used to see the whole production and stock points. Further information on topic can be found on 3.5 Value Stream Mapping

3.4.2 The Precedence Diagram

Afterwards, Precedence Diagram is used to map the precedential relationships of the tasks in the line. Further information on topic and precedence diagram for the subject of this study can be found on 3.6 Precedence Diagram

3.4.3 The Linear Model

After precedence Diagram with the inputs on the hand, the mixed integer linear programming model is structured.

At the beginning of this study, the mixed integer linear model is structured to achieve solution of the problem considered and desired outcomes. The model in below was made however this linear model does not allow the usage of some complicated constraints in mathematical model hence problem was not addressed with the model fully correct. This linear model lacked freedom and flexibility that was required, it limited possible outcomes so it was not a good fit.

Model can be seen in below;

$$\text{Min } Z = C \quad (1)$$

$$\sum_{j=1}^N X_{ij} = 1 \quad \forall i \in \text{set_I} \quad (2)$$

$$X_{ij} + X_{pn} \leq 1 \quad \forall i, j, p \in P_i, n \geq (j + 1) \quad (3)$$

$$\sum_{i=1}^M (X_{ij} \times t_i) \leq C \quad \forall i \in \text{set_I} \quad (4)$$

In the equations mentioned above, objective equation (1) is minimization of cycle time same. First constraint (2) ensures each task is assigned to at least and maximum one station. Second constraint (3) is constraint for precedence relationships for tasks. The last constraint (4) is cycle time constraint. It ensures that total processing time of all tasks assigned to a station is prevented from exceeding the cycle time.

3.4.4 The Non-Linear Model

Furthermore, to create a more advanced and realistic model and to give more freedom so non-linear constraints can be added, the mixed integer non-linear programming model is presented.

The linear model was not sufficient. Lack of non-linear constraints made the model simple, thus it did not satisfy the needs of analyzed company. These facts led to decision of making a non-linear model which led to finding more complicated constraints to reduce cycle time and to match the company's needs.

First improvement was adding operator assigning variable and constraints to mathematical model. As mentioned in, 3.3 Problem Definition, it was not possible for analyzed company to add another workstation due to distancing rule, company is following. So, idea of making a multi-manned workstation was emerged. Thought process behind it was making more balanced line using this two men station and the removed station can be used as side workstation for tasks without precedence which would lower the cycle time even further.

Another feature of the model is that it has precedence constraint for operators which gives substantial freedom and flexibility to itself. Company that was analyzed in the study did not have any preference between operators but in another scenario this constraint can be used to assign certain operators to certain workstations.

Second improvement was adding parallel task assignment. One of the main assumptions in traditional assembly line balancing problem is the serial arrangement of workstations. According to this assumption; each task can be assigned to only one station; it is not allowed to assign a task to more than one station or to create more than one workstation of equal nature. The assumption that stations are arranged serially in

assembly lines results in the minimum cycle time value to be limited to the duration of the task with maximum processing time. Therefore, it is one of the biggest disadvantages of this assumption that the tasks whose processing time is greater than the cycle time limit the production speed. The parallel task assignment approach based on task parallelism allows the line to be arranged in different ways. With this approach, a task is allowed to take place in more than one station. In this way, the line efficiency increases.

Due to the parallel task assignment approach, some tasks may take place at more than one station, provided that priority relations and the rules of non-division of business are satisfied. This approach was done by changing constraint related to cycle time and adding another decision variable (Y_i) and with the addition of parameter U number of parallel tasks are limited.

Final state of the non-linear model can be seen in 3.7 Mathematical Model and is explained in details.

3.5 Value Stream Mapping

In this study, the value stream mapping method was used to see the whole material and information flow from start to finish and the relationship between the information flow and the material flow. It was also used for the determination of precedence diagram and the development of suitable models, software, etc. in further methods. In addition, the value stream mapping study has made it easier to see and evaluate the data at the appropriate locations.

The data was obtained from the production planning and integrated business planning units. In addition, the study was conducted for a single product group.

In the figure shown below, the VSM containing the work area where the study is made, is given. The examined line is “Horizontal-line” and it is marked with a circle. If it is desired to examine in detail, it can be accessed from the attached “VSM” file.

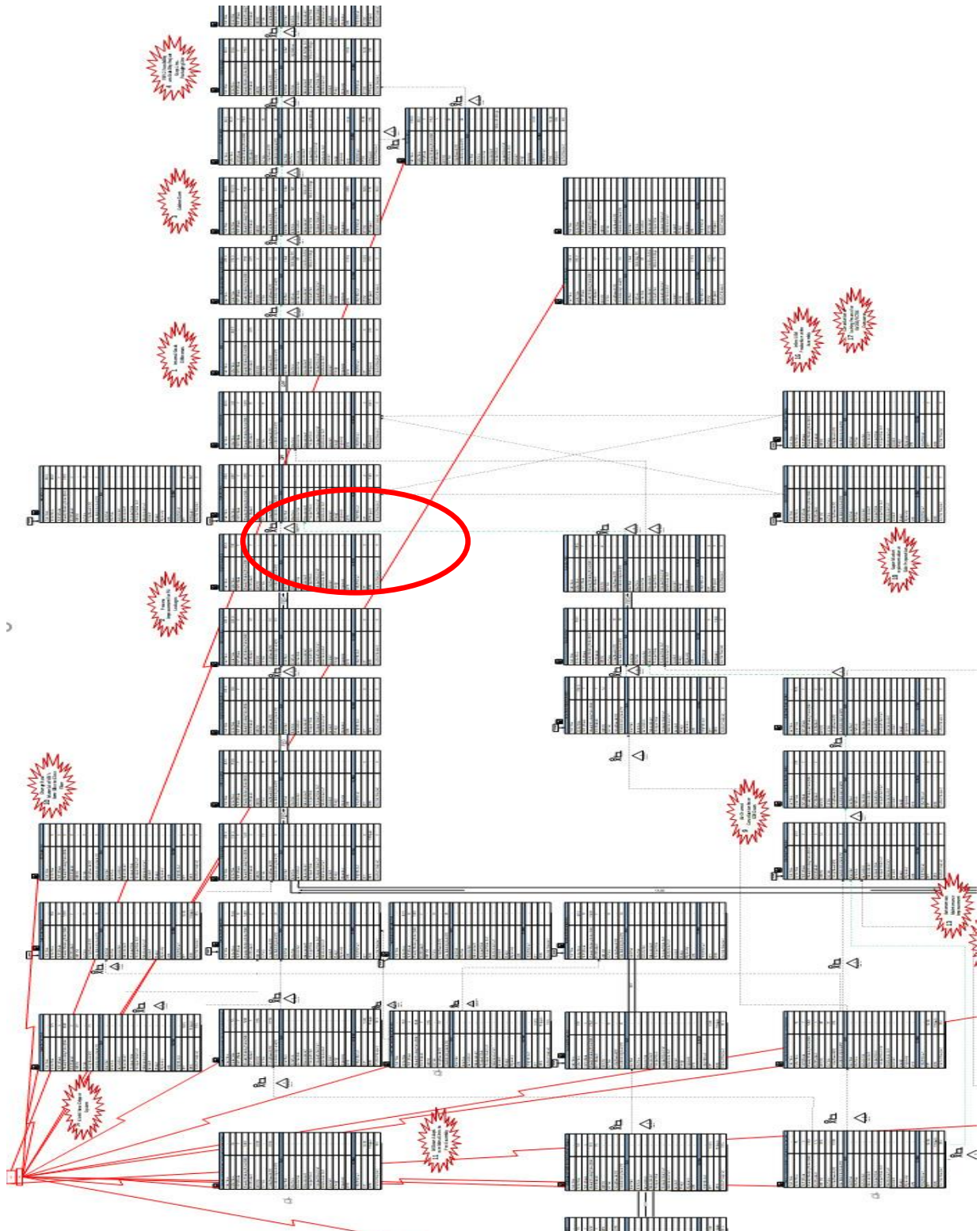


Figure 7:VSM

3.6 Precedence Diagram

In this section, precedence diagram can be seen below. This diagram is filled with total process times and tasks precedence. The Precedence Diagramming Method, with its activity-on-node network and additional precedence relationships, offers a simpler and increasingly adaptable system way to deal with demonstrating complex and large tasks.

Tasks precedence diagram can be seen below;

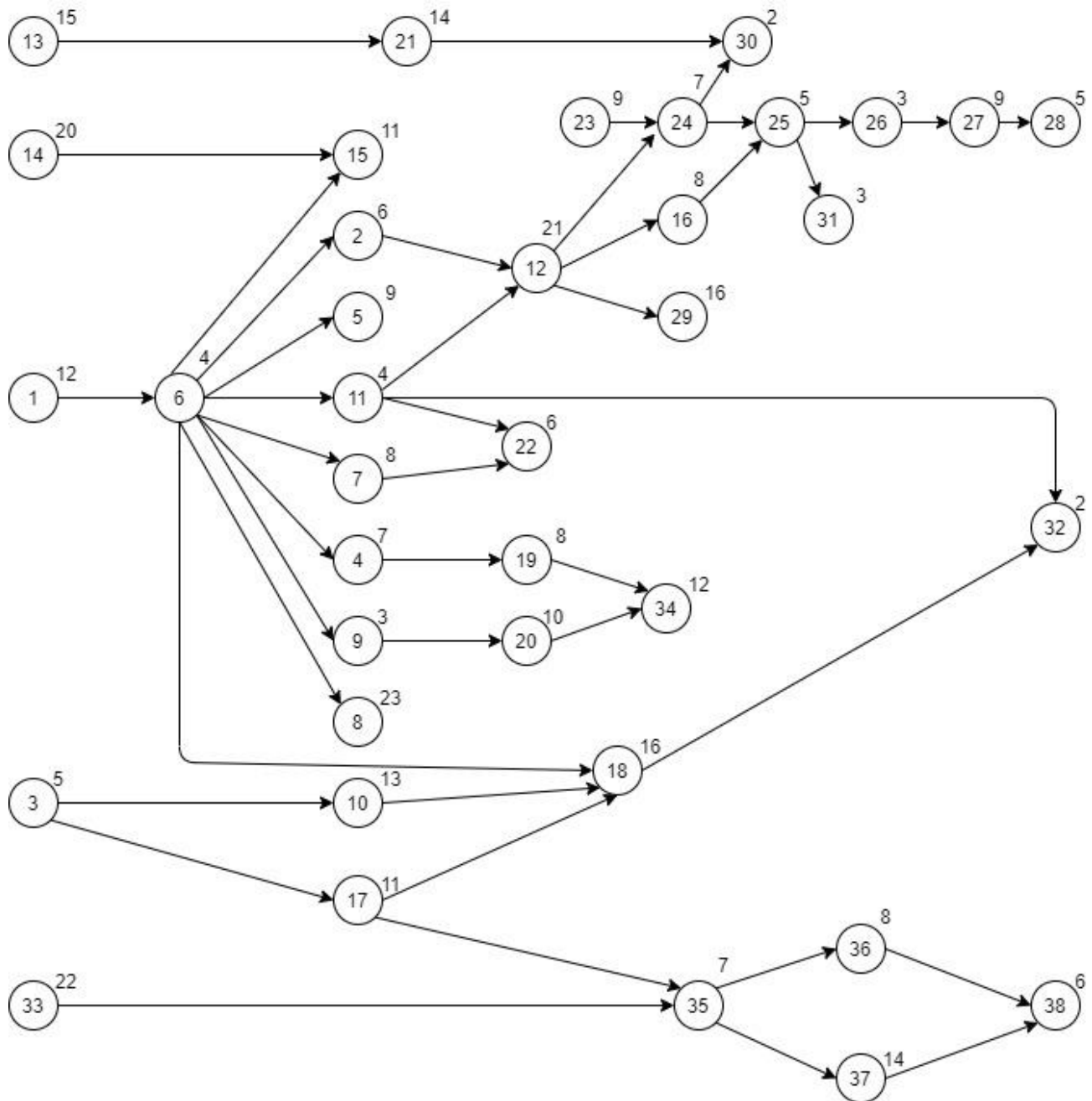


Figure 8: Precedence Diagram

The Precedence Diagramming Method (PDM) is the most widely-recognized strategy to draw network diagrams. Arrow means that task must be completed before starting the other one. If two or more arrows are targeting a task, that means all of them must be completed to start that task.

In the diagram mentioned above, activities are designated as 1, 2, 3, 4, 5, 6, ..., 37,38. Each task's process times are the numbers written above the boxes.

Task number 1, 3, 13, 14, 33 does not have a precedence before them so they can be done in any workstation. Other tasks have precedence. As an example, task 6 is a predecessor task that logically comes before a dependent tasks with numbers 2, 4, 5, 7, 8, 9, 11, 15 in this precedence diagram. Task 12 is a successor task and a dependent task. It inevitably occurs after the task 2 and task 11 on top of that it is a processor task that comes before tasks 16, 24 and 29.

3.7 The Mathematical Model

This model focuses on plants with single model assembly lines. There are linear workstations that have a number of operators assigned to them. The aim of this model is allocating tasks to ensure balance between workstations and achieving the desired cycle time under the consideration of several constraints.

3.7.1 The Definition of the Problem

In a horizontal assembly line, defined tasks for similar products are assigned to workstations where operators working are also assigned with the model. However, while making these assignments following should be taken into consideration;

- Tasks have different precedences. These precedences determine the sequence of the tasks.
- If the workstations are not balanced as desired, as result of this situation it's mean the operators do not have equal amount of workload.

- Capacity increase through additional workstations is not allowed due to location of this assembly line is not enough for another workstation.
- There are time wastes while transferring parts between operators in workstations.
- High cycle time prevents from achieving requested capacity increase.

3.7.2 The Assumptions

In addition, several assumptions are made within the scope of the study. However, these are made by taking expert opinions to market them more realistic.

Assumption No	Assumption Description
1	Equipments and tools are always on the right spot and on the right condition.
2	Processing times are common and stable for all tasks.
3	Calculation unit of processing time is second.
4	Operators have 100% worker efficiency. Fatigue and other allowances are not included.
5	Distances between workstations are set considering operators to move sufficiently.
6	The task assigned to the workstation must be completed.
7	There is no pause in assembly line.
8	Material scrap rates are not included in the study.
9	Maximum two workers can work in a workstation.

10	Tasks cannot be divided, However, some tasks can be assigned to more than one station. For mathematical model proposed in this study a task can take place at most two stations
11	Equipments and equipment costs are not considered for any task assigned to more than one station.

Table 1:Assumption

3.7.3 The Sets and Indices

Table that contains sets and indices can be seen below;

Set	Meaning
I	Set of work i , $i= \{1,...,M\}$
J	Set of station j , $j= \{1,...,N\}$
K	Set of operators k , $k= \{1,...,O\}$
P	Set of precedense for task p , $p= \{1,...,M\}$
L	Set of presedence for operator l , $l=\{1,...,O\}$

Table 2:The Sets and Indices

3.7.4 The Parameters

Table that contains parameters and their meanings can be seen below;

Parameter	Meaning
M	Number of task
N	Number of station
O	Number of operatör

t_i	Processing time of task i
P_i	Presedence of task i
L_k	Presedence of operator k
U	Upper-limit for parallel tasks
R	Upper-limit for multi-manned stations

Table 3: The Parameters

3.7.5 The Decision Variables

Table that contain decision variables can be seen below;

Variable	Meaning
X_{ij}	$\begin{cases} 1, \text{ If task } i \text{ is assigned to station } j \\ 0, \text{ otherwise} \end{cases}$
Y_i	$\begin{cases} 1, \text{ If task } i \text{ is assigned to multiple stations} \\ 0, \text{ otherwise} \end{cases}$
W_{kj}	$\begin{cases} 1, \text{ If operator } k \text{ is assigned to station } j \\ 0, \text{ otherwise} \end{cases}$
V_j	$\begin{cases} 1, \text{ If multiple operators are assigned to station } j \\ 0, \text{ otherwise} \end{cases}$
C	Cycle time

Table 4: The Decision Variables

3.7.6 Data

The related data that used for this study can be seen below;

Station Number	
1	5
2	6
3	7
4	

Table 6: The workstations

Operator Number	
1	5
2	6
3	7
4	8

Table 5: The operators

Each task time and task's precedence can be seen below;

Task No	Task Time(sn)	Precedence	Task No	Task Time(sn)	Precedence
1	12	()	20	10	9
2	6	6	21	14	13
3	5	()	22	6	7,11
4	7	6	23	9	()
5	9	6	24	7	23
6	4	1	25	5	16,24
7	8	6	26	3	25
8	24	6	27	9	26
9	3	6	28	5	27
10	13	3	29	16	12

11	4	6	30	2	21,24
12	21	(2,11)	31	3	25
13	15	()	32	2	11,18
14	20	()	33	22	()
15	11	(6,14)	34	12	19,20
16	8	12	35	7	17,33
17	11	3	36	8	35
18	16	(6,10,17)	37	14	35
19	8	4	38	6	36,37

Table 7: The Tasks

3.7.7 Mathematical Model

We formulated the problem as a mixed-integer linear programming (MILP) model with the main objective of total completion times as given below:

$$\text{Min } Z = C \quad (1)$$

$$\sum_{j=1}^N X_{ij} \geq 1 \quad \forall i \in \text{set}_I \quad (2)$$

$$\sum_{j=1}^N X_{ij} \leq 2 \quad \forall i \in \text{set}_I \quad (3)$$

$$\sum_{k=1}^o W_{kj} \geq 1 \quad \forall j \in set_I \quad (4)$$

$$\sum_{k=1}^o W_{kj} \leq 2 \quad \forall j \in set_I \quad (5)$$

$$\sum_{j=1}^N W_{kj} = 1 \quad \forall k \in set_K \quad (6)$$

$$X_{ij} + X_{pn} \leq 1 \quad \forall i, j, p \in P_i, n \geq (j + 1) \quad (7)$$

$$W_{kj} + W_{ln} \leq 1 \quad \forall k, j, l \in L_i, n \geq (j + 1) \quad (8)$$

$$\sum_{j=1}^N X_{ij} = Y_j + 1 \quad \forall i \in set_I \quad (9)$$

$$\sum_{i=1}^M ((X_{ij} \times Y_i) \times t_i \times 0.5 + ((X_{ij} - (X_{ij} \times Y_i)) \times t_i)) \div$$

$$\sum_{k=1}^o (W_{kj}) \leq C \quad \forall j \in set_I \quad (10)$$

$$\sum_{j=1}^N Y_i \leq U \quad (11)$$

$$\sum_{k=1}^N W_{kj} = V_j + 1 \quad \forall j \in \text{set_I} \quad (12)$$

$$\sum_{j=1}^N V_j \leq R \quad (13)$$

$$X_{ij}, Y_i, W_{kj}, V_j \in \{0,1\} \quad (14)$$

$$C \geq 0 \quad (15)$$

Object function of this mathematical model (1) has a goal of minimizing cycle time. This inequation (2) is constraint for assignment of tasks. It ensures they sat least get assigned to a station. Next inequation (3) is also for assignment of tasks. Tasks should get assigned to utmost two different situation. Constraint (4) makes sure at least an operator gets assigned to each station. (5) Makes sure utmost two operators get assigned each station. With constraint (6) each operator is assigned to a station. For p is precedence of task i and element of P_i (7), this constraint ensures tasks are assigned in regard to precedence diagram. For l is precedence of operator k and element of L_k, this constraint (8) ensures tasks are assigned in regard to precedence for operators. (9) It is used for connecting decision variables. (10) It is the cycle time constraint. It ensures that total processing time of all tasks assigned to a station is prevented from exceeding the cycle time with the consideration of how many operators are assigned to that station. (11) It limits number of parallel tasks with upper limit parameter: U. (12) It is used for connecting decision variables. (13) It limits number of multi-manned stations with upper limit parameter: R. (14) Binary variable. (15) Positive variable.

4. RESULTS AND CONCLUSION

CHAPTER IV

4.1 Model Results and Analyzes

In this section the mathematical model mentioned before is applied to the data set obtained from studied company and results are analyzed. Also programming language that was used for this goal, is described briefly. All the information is supported with appropriate tables.

4.1.1 Python

The mathematical model is solved in python programming language. Pyscipopt, the interface library for SCIP Optimization Suite was used to achieve the solution, and Pandas library was used to transfer the data and to visualize the results. Python was chosen because of large number of libraries it offers and its ability to solve complex problems quickly and effectively.

4.1.2 Result Tables

Results of code that solved the mathematical model can be seen below. This table shows which tasks and which operators are assigned to which station. Parallel assigned tasks are marked in red color.

Workstations	Tasks assigned	Operators assigned	Total time
1	1, 9, 6, 33, 3	Op.no:1	46
2	10, 4, 20, 19, 7	Op.no:2	46

3	17, <u>16</u> , 12, 11, 2	Op.no:3	46
4	18, 34, 29	Op.no:4	44
5	5, 14, 23, 24	Op.no:5	45
6	32, 25, 26, <u>16</u> , 15, 27, <u>8</u>	Op.no:6	45
7	36, 35, 37, <u>8</u> , 14, 21, 22, 28, 30, 31, 38	Op.no:7 and Op.no:8	97/2= <u>48.5</u>
			Objective: 48.5

Table 8: Results-1

4.2 Sensitivity Analysis

In this section, 2 sensitivity analyses are proposed. These analyses are related with number of tasks and workstations and parallel task assignment. With these analysis, Mathematical Model is tested in different scenarios.

4.2.1 Sensitivity Analysis-1

Model is tested in with the parallel task assignment and multi-manned station constraints and variables are removed. This results in cycle time increasing to 53 seconds. Results for this scenario can be seen below;

Workstations	Tasks assigned	Total time				
1	1, 3, 6, 14, 17	52	5	13, 25, 26, 27, 16, 34	51	
2	35, 33, 7, 37	51	6	18, 36, 21, 10, 30	53	
3	2, 8, 9, 23, 20	52	7	5, 28, 31, 29, 15, 32, 38	52	
4	24, 22, 19, 12, 11, 4	53			Objective:	53

Table 9: Results-2

Importance of parallel task assignment can be seen with these results. Not being able to assign tasks with high processing time, such as task 8, task 12, task 33, to two workstations instead of one results in less efficient line.

4.2.2 Sensitivity Analysis-2

With the number of workstations decreased to 5, number of operators decreased to 6 and number of tasks decreased to 12, how well mathematical model of this study does in different situations can be seen in a more clear way. In this scenario, objective cycle time is 19,5 seconds. Results for this scenario can be seen below;

Workstations	Tasks assigned	Operators assigned	Total time
1	1, 6, 9	Op.no:1	19
2	<u>10</u> , 7, 3	Op.no:2	19,5
3	4, 2, <u>10</u>	Op.no:3	19,5
4	8, 11, <u>12</u>	Op.no:4, Op.no:5	38.5/2=19,25
5	5, <u>12</u>	Op.no:6	19,5
Objective:			19,5

Table 10: Results-3

These analysis shows the mathematical model gives the best result possible even if parameters are changed.

4.3 Discussion

This study was made to illustrate a methodological approach by creating and applying a mathematical model. The main goal was to analyze certain line in certain plant of home appliances manufacturing company and to balance this line by reducing its cycle time. Hence the mathematical model was created. Although this mathematical model does well with dataset on hand for this study, still while applying it to the dataset some problems were noticed.

Problems occur when data gets bigger. Trying to run the model with a data set containing more work-stations and tasks, after coding mathematical model in python, results in the program's decoding time going to hours, days or even the result may not be achieved depending on the situation. Such reasons are the limitations of the mixed

integer model. These restrictions can be overcome with the heuristic approach. This study can be a guide for the transition to the meta-heuristic approach. With the meta-heuristic approach, new constraints and variables can be added and the complex problems that are desired can be solved as a result.

Suggested Meta-Heuristic approach is Genetic Algorithm(GA). Because with Genetic Algorithm parameters such as workers and tasks can be easily be represented with gens. In general, the power of Genetic Algorithm comes from the fact that the technique is robust and can deal with a wide range of problem areas. Although Genetic Algorithm is not guaranteed to find the optimal solution, they generally give good enough solutions with reasonable computational requirements.

4.4 Conclusion

In this study, a mathematical model is proposed for assembly line of a plant with the objective of reaching desired production capacity by lowering its cycle time. Problem on hand was identified as Simple Assembly Line Balancing Problem Type-2(SALBP-2). Firstly, a linear model was developed but after not being satisfied with the results for linear model, non-linear mixed integer programming was used.

As for non-linear constraints, model adds parallel task assignment and operator assignment to a generic assembly line balancing model. This way, efficiency of line gets increased and cycle time goes even lower. Furthermore, it adds much more flexibility to the model, making it viable in various scenarios.

This model was tested with datasets provided from studied company. Due to company's policy changes were made on parameters. The results from actual dataset and sensitivity analyses showed that the model can be used to balance an assembly line and to reduce its cycle time.

As a result of this study, a new non-linear mixed integer mathematical programming has been introduced to the literature for the solution of SALBP-2 by clearly demonstrating the gains of the parallel line assignment and multi-manned workstation approach in line balancing studies. Significant gains have been shown in terms of efficient use of production time and minimization of idle times. In the following studies, this model can

be improved by transforming it to a Genetic Algorithm from Meta-Heuristics Approach to use it in even more complicated, wide-ranging industry application.

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Acknowledgement 2

